

Emirhan Şimşek

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Education

Boğaziçi University | Istanbul, Turkey

Computer Engineering & Industrial Engineering (Double Major) | 01/2027

GPA: 3.37/4.00

Technical University of Munich (Erasmus Exchange Program)

Informatik (Computer Science) | 08/2024

Introduction to Deep Learning, Augmented Reality, German B2.1, Entrepreneurship

Hüseyin Avni Sözen Anatolian High School

GPA: 97.44/100

Technical Skills

Python, Java, ML Models, OOP, Data Structures & Algorithms, R, Statistics, SQL, Excel

Languages

English(C2), German(B1-B2), Turkish(Native)

Skills

- Collaboration skills
- Ability to focus intensely on a task for hours
- Ability to learn and improve fast to achieve a given task

Projects

2026 Jan – Research at Boğaziçi University about VRP with Time Intervals

- Developed mathematical models for a Vehicle Routing Problem (VRP) with Time Intervals.
- Wrote a code that solves the model.
- Aiming to publish an article at an international conference.

2025 Aug – Next Node Prediction in an Internship at ICRON for an Internal Tool of the Company

- Used Transformers, did next word prediction, and used majority voting.
- Used many embedders like Bert, FastText, and minilm.
- Did predictions using cosine similarity of the embedded vectors.

2025 Jun – Research at Koç IUI Lab about Multimodal Speech Sketch Interfaces

- Built an interface that takes user input as sketch and voice commands, and then animates the sketch.
- Was my first real research experience. I read many academic papers.

2025 Jan – Multiple Time Series Forecasting During an Internship at ICRON (with Python Darts Library)

- Used NBeats and NHits models to do 556 forecasts on 556 time series.
- Did data preprocessing. Turned the data in icdb form to the suitable form. First, turned the data into CSV, then to Python Pandas DataFrame, and then to Darts TimeSeries format. Merged the tables. Did scaling.
- Did Hyperparameter Tuning.
- Presented the steps followed and explained the decisions taken and conclusions made (such as observing overfitting and how it was mitigated).
- Learned well how to work with time series data.
- Learned how to save trained models to use them in future prediction making.

2024 – Deep Learning course at Technical University of Munich (with Python)

- Developed neural networks manually using NumPy. Defined the functions by hand, built and trained the model, and made predictions. Building by hand enabled me to understand how a neural network works.
- Used PyTorch.
- Learned and applied Data Preprocessing, Model Parameter Initialization, Hyperparameter Tuning, and other ML concepts.
- Developed and utilized neural networks, including Feature Extractor + Classifier pair for NLP, Autoencoders, and CNNs. Additionally, learned well about RNNs, LSTMs, Attention, and Transformers. Also, touched upon GNNs, Variational Autoencoders, GANs, Diffusion Models, etc.
- Did "Facial Keypoint Detection" and "Semantic Segmentation" using CNNs.

2024 – Data Structures and Algorithms with Java

- An AVL tree that ranks and traverses nodes which represents people with various attribute values.
- A hash map that allocates employees to positions by input values and commands while adhering to initial constraints.
- A music shuffle programme that sorts songs by their scores while adhering to initial constraints. I used heaps.
- These projects run thousands of lines of input thus required time complexity analysis. The consideration of time complexity enabled run time to be below one minute instead of hours.

2023 – Object-Oriented Programming with Java, Applying Algorithms

- A programme that finds routes in Istanbul metro lines. I used depth first search algorithm and visualised the path.
- A game similar to Bubble Trouble. I used inheritance.
- A programme that fills lakes and labels them. I used breadth first search algorithm.

2022 – Program to turn a smart phone into a pen

- Developed a program that receives data from smart phone sensors, outputs the written words, and visualizes the movement of the phone.
- I turned the project into an app and published it. The app can also recognise words.

2021 – Research on systems of product categorization

- Suggested a new method of categorizing and listing products in a report and put forward a website design.
- Reviewed UN's "International Standard Industrial Classification of All Economic Activities Revision 4"(p.96-160). Examined EU product standards. Read the book "Managing quality: the strategic and competitive edge" by David A. Garvin.

2018 – Physics project regarding climate crisis, organic waste management, composting

- Modelled a waste management system and designed composters for urban and rural composting.
- Did a SolidWorks Flow simulation to prove that the air flow I suggested is possible through a manifold.

2017 – Remote control car with Arduino

- Built a car with Arduino. Controlled it with my smartphone's Bluetooth. Made me realize that I like building things.